

Winner Takes All in **Competitive Bottlenecks**

Lester T. Chan

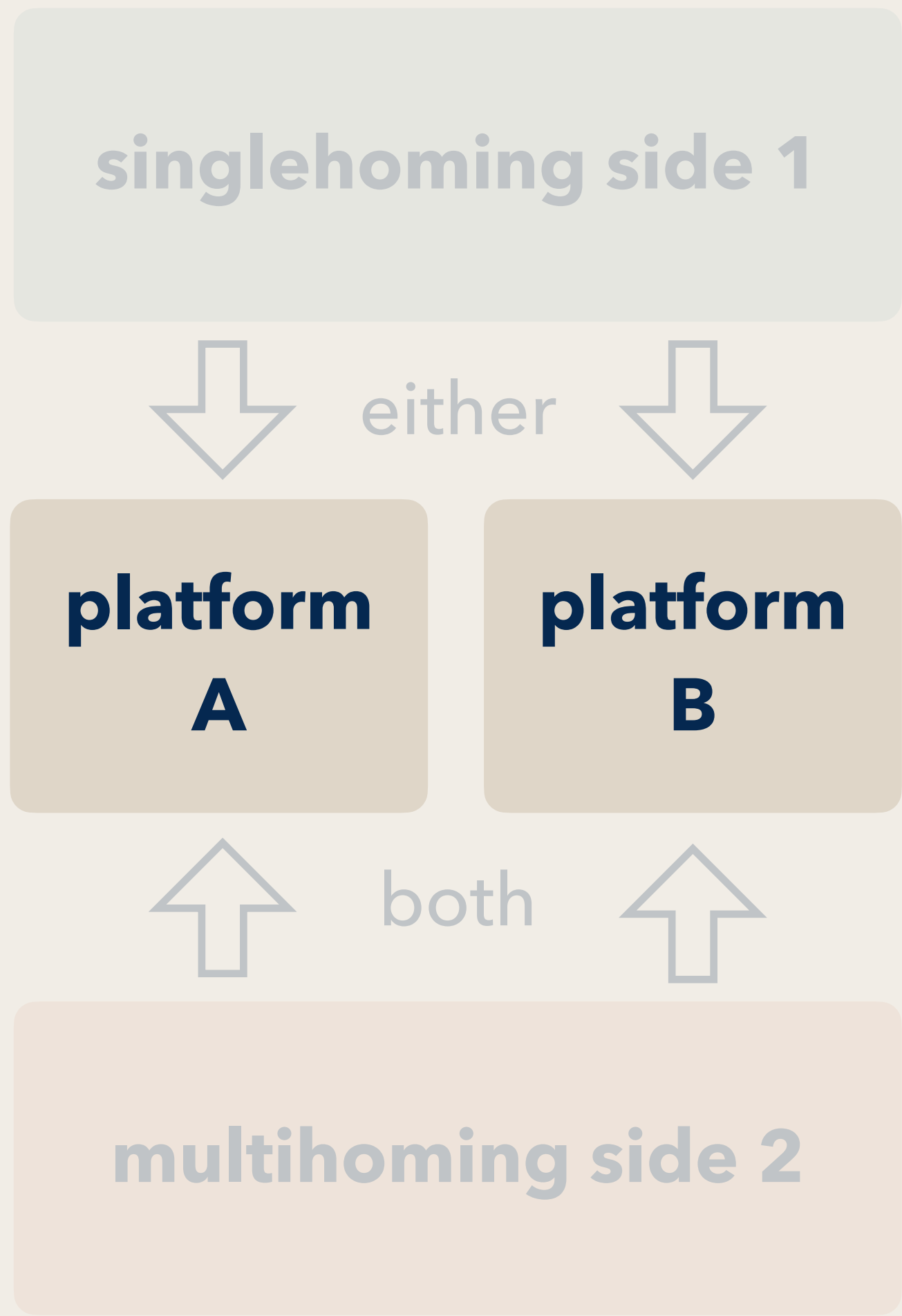
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Competitive Bottlenecks

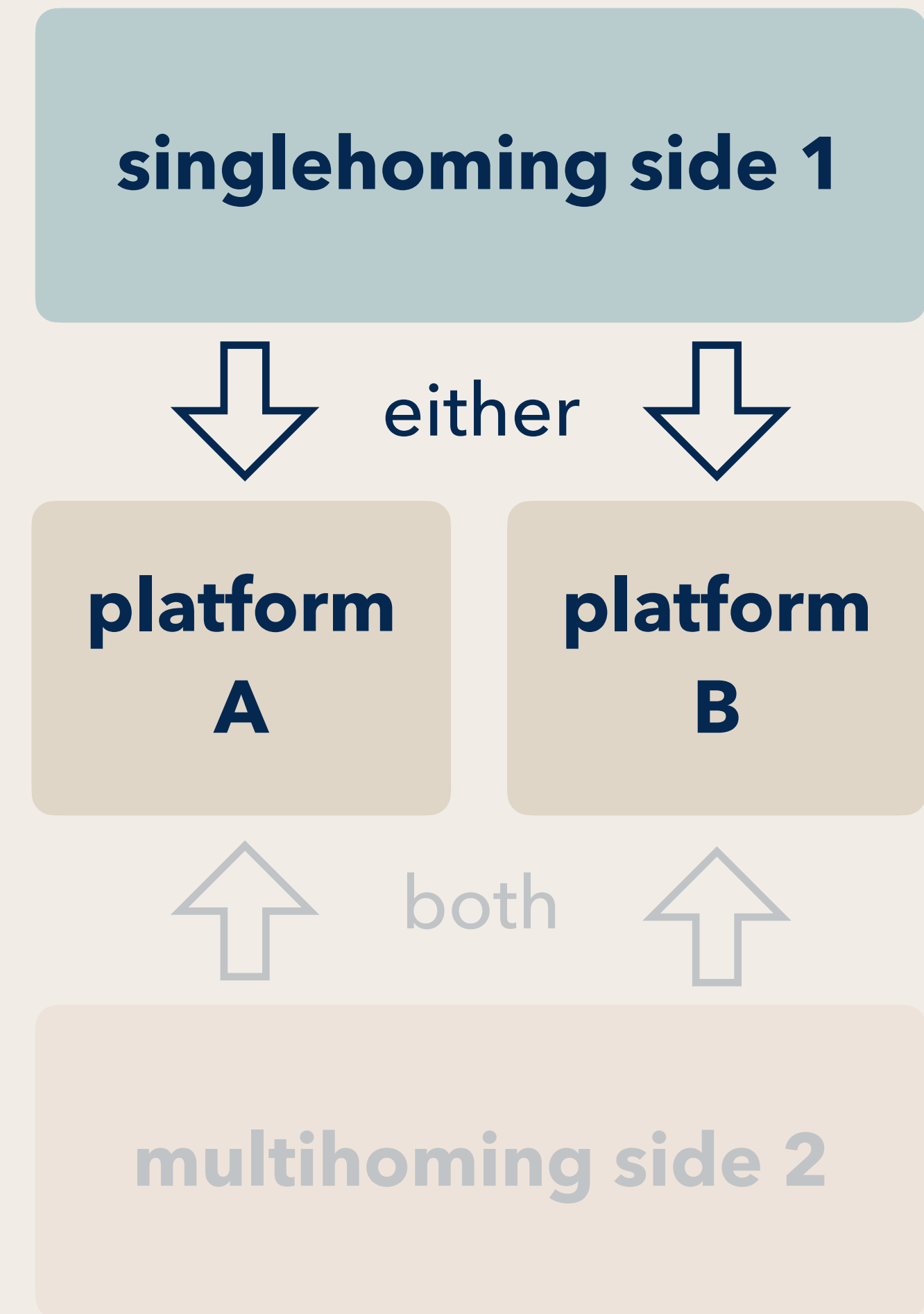
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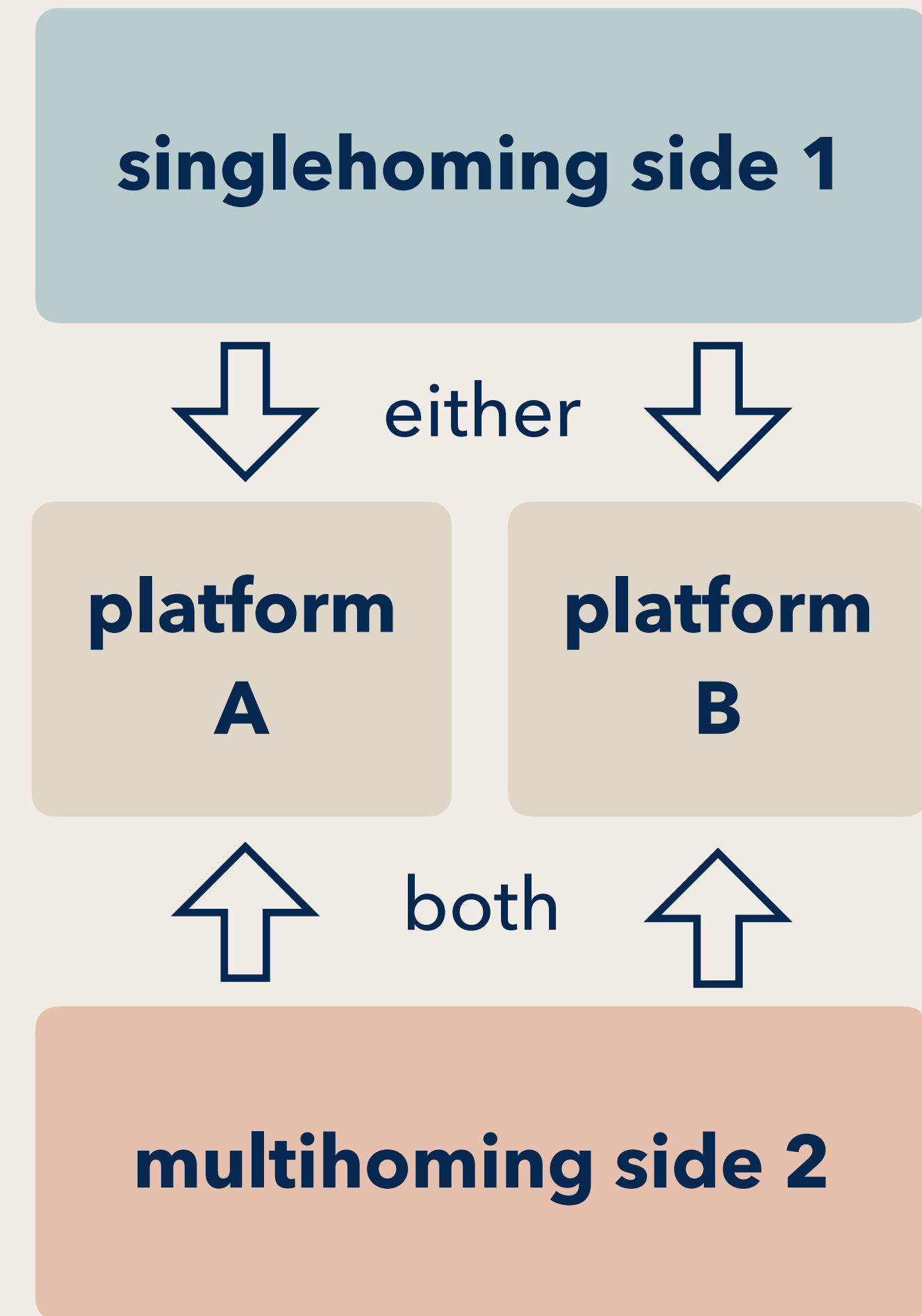
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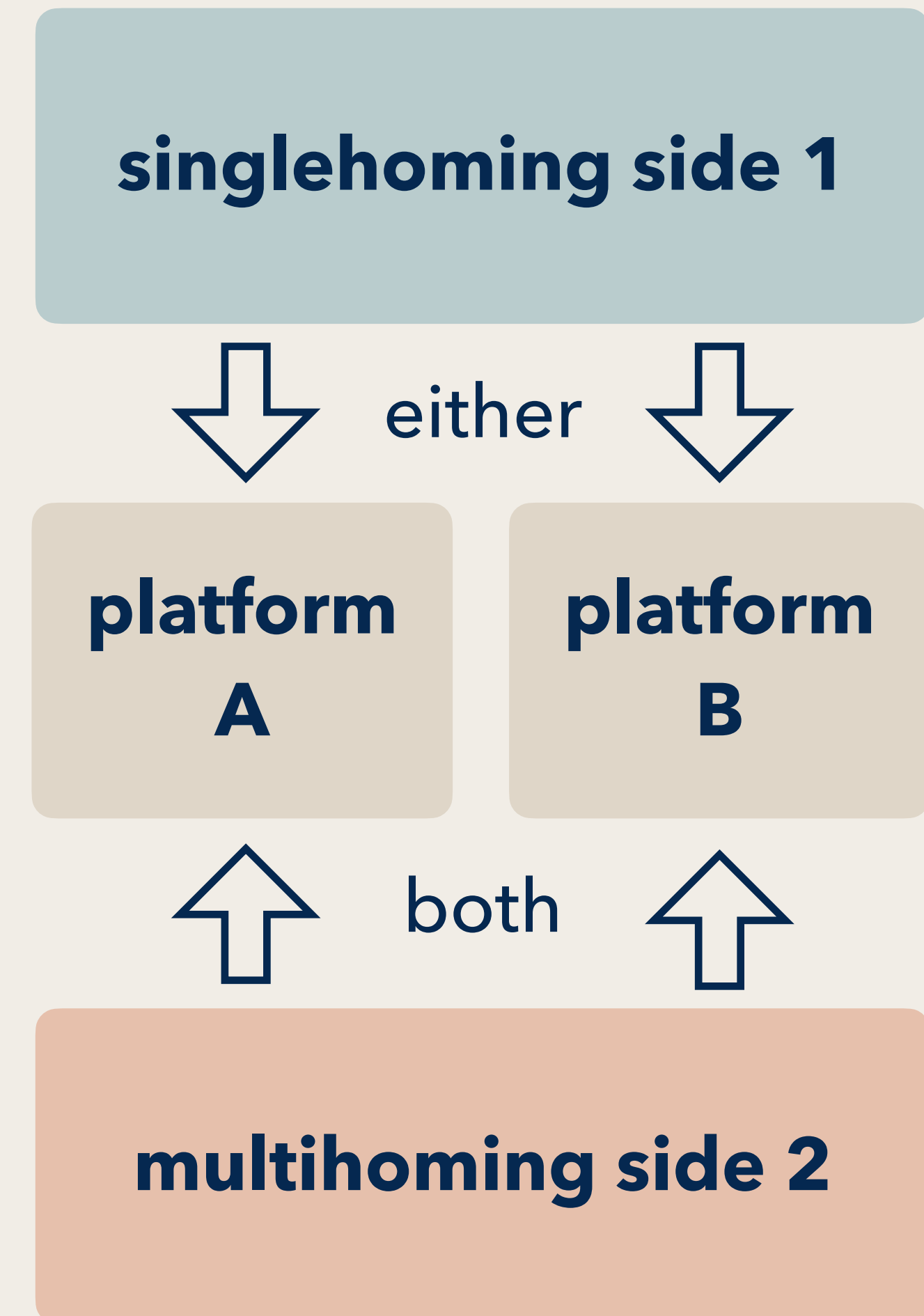
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 - users typically settle on a single OS
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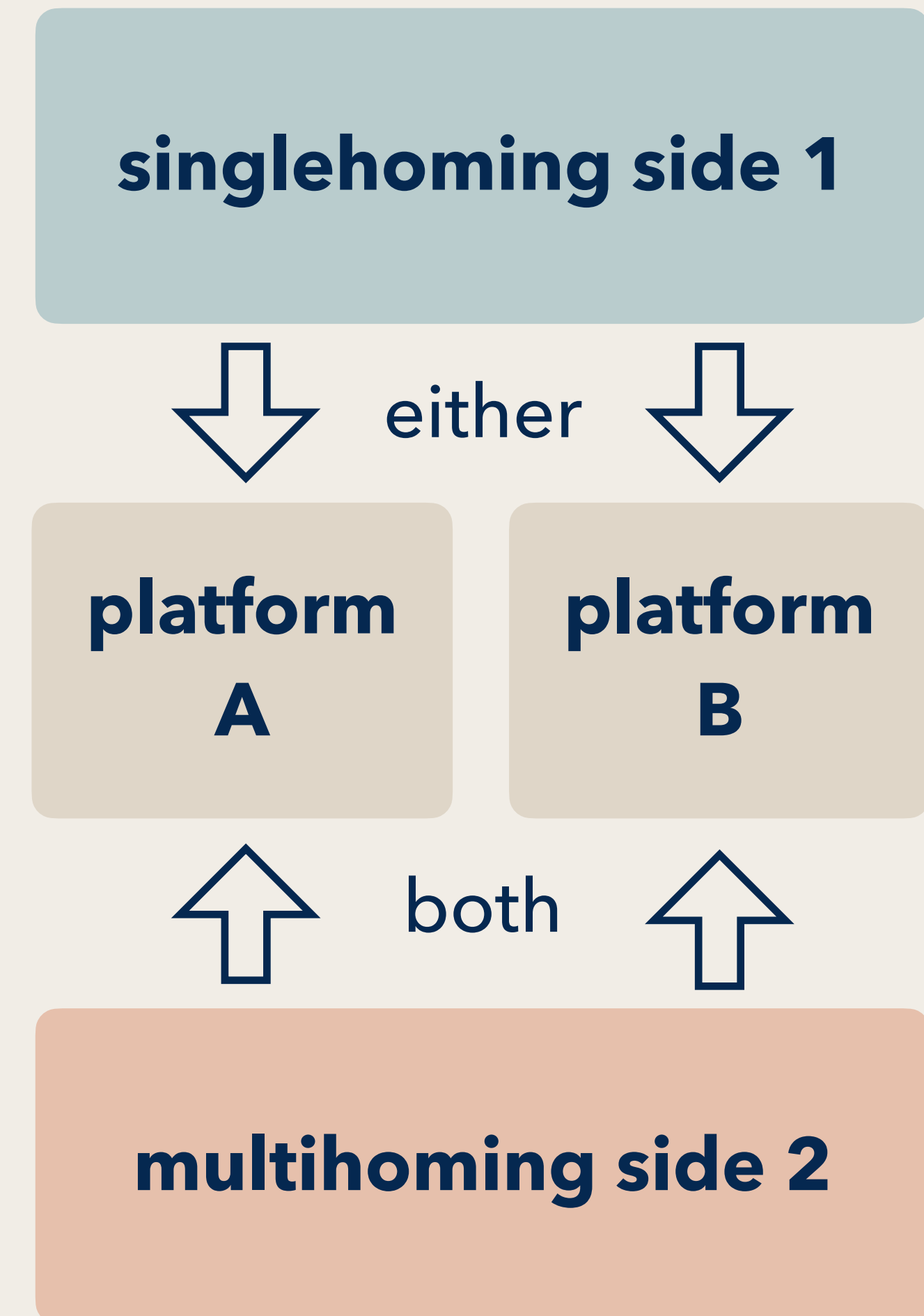
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Examples

- operating systems
 - users typically settle on a single OS
 - developers offer software for multiple OS
- game consoles
 - users typically own one console
 - developers release the same title for multiple consoles



How Should Platforms Price the Two Sides?

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in software markets, platforms often:

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 - a well-established equilibrium refinement

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Results

- the “chicken-&-egg” participation game is a **best-response potential game**
 - a well-established equilibrium refinement
- the refinement uniquely selects a **winner-take-all equilibrium**
 - the equilibrium pricing rationalizes the Evans–Schmalensee puzzle

Model

Equilibrium Analysis without Refinement

Equilibrium Analysis with Refinement

Platform Design: Exclusive Offers?

if time permits

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Two-Stage Game: Timeline

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Stage 1: platforms A and B simultaneously choose prices

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- side-1 agent chooses A only, B only, or neither

$$\mathbf{n}_1 = (n_1^\emptyset, n_1^A, n_1^B, n_1^{AB} = 0)$$

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 n_i^k = #side- i agents joining k only
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- side-2 agent chooses A only, B only, both, or neither

$$\mathbf{n}_2 = (n_2^\emptyset, n_2^A, n_2^B, n_2^{AB} \geq 0)$$

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Side-1 payoff

$$u_1^k(\mathbf{n}_2, \mathbf{p}) = v_1^k (n_2^k + n_2^{AB}) - p_1^k$$

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Side-2 payoff

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$$u_2^{AB}(\mathbf{n}_1, \mathbf{p}) = u_2^A(\mathbf{n}_1, \mathbf{p}) + u_2^B(\mathbf{n}_1, \mathbf{p})$$

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key differences from relevant models

- **agents are homogeneous**
 - ▶ $\neq$ Hotelling-type model
- **platforms are asymmetric**
 - ▶ vertical differentiation

stage-2 game admits **multiple equilibria**

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 - platforms never set any prices that bring negative profits
- monotonicity: price $\mathbf{p}^k = (p_1^k, p_2^k)$ weakly $\downarrow \Rightarrow$ #participants $(n_1^k, n_2^k + n_2^{AB})$ weakly $\uparrow$
 - Caillaud-Jullien 03; Armstrong-Wright 07

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Two Classes of Equilibria

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Market-sharing equilibrium: side-1 agents split and side-2 agents join both

- side-1 agents split: $n_1^A \geq 1$ and $n_1^B \geq 1$
- side-2 agents join both: $n_2^{AB} = N_2$

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- ▶ B's deviation profit $p_1^A N_1 - (v_1^A - v_2^B)N_1 N_2 \leq 0 \Rightarrow$ upper bound on price $p_1^A \leq (v_1^A - v_2^B)N_2$

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the winner identify depends on both sides;
it is **not** determined by the single homing side

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- side-1 agents split between A and B (with side-2 agent multihoming)
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 - if platform k chose $p_1^k > 0$ then it could attract side-1 agents by cutting its own price

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 - if platform k chose $p_1^k > 0$ then it could attract side-1 agents by cutting its own price
 - if platform k chose $p_1^k < -v_2^k N_2$ then the subsidy paid to side 1 would $>$ the max revenue from side 2

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- side-1 agents split between A and B (with side-2 agent multihoming)
 - side-1 agents must be indifferent: $v_1^A N_2 - p_1^A = v_1^B N_2 - p_1^B$
- price bound $p_1^k \in [-v_2^k N_2, 0]$
 - if platform k chose $p_1^k > 0$ then it could attract side-1 agents by cutting its own price
 - if platform k chose $p_1^k < -v_2^k N_2$ then the subsidy paid to side 1 would $>$ the max revenue from side 2
- combining these bounds with the indifference condition gives " $\Rightarrow$ "

$$(v_1^A - v_1^B) N_2 = p_1^A - p_1^B \in [-v_2^A N_2, v_2^B N_2]$$

Equilibrium Multiplicity

These existence conditions can overlap

- the same parameter may support multiple equilibrium outcomes

equilibrium	
A-tipping	$v_1^A + v_2^A \geq \max\{v_1^B, v_2^B\}$
market-sharing	$-v_2^A \leq v_1^A - v_1^B \leq v_2^B$

the market may tip to an inferior platform

- there can exist an A-tipping equilibrium even if A is inferior: $v_1^A < v_1^B$ $v_2^A < v_2^B$

the conventional wisdom applies to this case

- subsidize side 1: $p_1^A \leq 0$ $p_1^B \leq 0$
- monetize side 2: $p_2^A \geq 0$ $p_2^B \geq 0$

Model

Equilibrium Analysis without Refinement

Equilibrium Analysis with Refinement

Platform Design: Exclusive Offers?

if time permits

Model

Equilibrium Analysis without Refinement

Equilibrium Analysis with Refinement

Platform Design: Exclusive Offers?  if time permits

Best-Response Potential Monderer-Shapley 96; Morris-Ui 04, 05

A finite normal-form game $G = \langle I, (A_i, u_i) \rangle$ is a **best-response potential game** if

- $\exists$ potential function $\Phi : A \rightarrow \mathbb{R}$ that preserves best responses

$$\arg \max_{a_i} u_i(a_i, \lambda_{-i}) = \arg \max_{a_i} \Phi(a_i, \lambda_{-i}) \quad \forall i \in I \quad \forall \lambda_{-i} \in \Delta(A_{-i})$$

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potential game if $\exists \Phi : A \rightarrow \mathbb{R}$ that preserves payoff differences
 $u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i}) = \Phi(a'_i, a_{-i}) - \Phi(a_i, a_{-i}) \quad \forall i \in I \quad \forall a_{-i} \in A_{-i}$

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Potential approach in recent IO

- two-sided markets $\neq$ competitive bottlenecks
 - Chan 21 and Hoshino–Panics 26

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Equilibrium Refinement: Potential Maximization

Equilibrium refinement

- potential-maximizing action profile $\arg \max_a \Phi(a)$ is generically a unique NE

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- robustness is a desirable property
 - agents may not know other agents' valuations
 - robustness examines which outcome survives that doubt

Equilibrium Refinement: Potential Maximization

Theorem: for a generic best-response potential game with no actions are strictly dominated, potential maximizers are **independent** of the choice of potential:

$$\arg \max_a \Phi(a) = \arg \max_a \Phi'(a) \quad \forall \text{ potentials } \Phi \text{ and } \Phi'$$

Equilibrium refinement

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Competitive bottleneck with $N_1 = N_2 = 1$

both subsidize side 1

- values $(v_1^A, v_2^A, v_1^B, v_2^B) = (4, 2, 2, 4)$ with prices $\mathbf{p}^A = \mathbf{p}^B = (-1, 1)$
 - joining neither is iteratively strictly dominated

payoff	A	AB	B
A	5, 1	5, 0	1, -1
B	1, -1	3, 2	3, 3

potential	A	AB	B
A	3	2.5	0.5
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Example

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these tables do not share the same payoff difference but the same best responses

$$BR_2 = \begin{cases} A & \text{if } \Pr(1 \text{ joins } A) \in (3/4, 1) \\ AB & \text{if } \Pr(1 \text{ joins } A) \in (1/2, 3/4) \\ B & \text{if } \Pr(1 \text{ joins } A) \in (0, 1/2) \end{cases}$$

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- potential-maximizing NE: (A, A)
 - A-tipping equilibrium in this subgame

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Theorem: Competitive Bottleneck Is Best-Response Potential Game

Given any price $\mathbf{p} = (\mathbf{p}^A, \mathbf{p}^B)$ the stage-2 subgame is a best-response potential game:

$$\Phi_{\mathbf{p}}(\mathbf{n}) = \sum_k \left[v_1^k n_1^k \left(n_2^k + n_2^{AB} \right) - p_1^k n_1^k - \frac{v_1^k}{v_2^k} p_2^k \left(n_2^k + n_2^{AB} \right) \right]$$

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interaction value
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Potential $\Phi_{\mathbf{p}}$

- ratio v_1^k / v_2^k serves as an exchange rate converting side-2 prices into side-1 payoffs

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Potential $\Phi_{\mathbf{p}}$

- ratio v_1^k / v_2^k serves as an exchange rate converting side-2 prices into side-1 payoffs
- construction idea
 - side 1: keep the original payoff structure
 - side 2: break the payoff structure, while respecting **piecewise linearity** to preserve BR

	without refinement	with refinement
equilibrium	market-tipping equilibria market-sharing equilibria	a unique equilibrium: market-tipping equilibria
winner	either platform may win: even an inferior one $\exists$ k-tipping: $v_1^k + v_2^k \geq \max\{v_1^{-k}, v_2^{-k}\}$	a unique winner: higher $\max\{v_1^k, v_2^k\}$ tipping to the inferior is ruled out
pricing	a continuum of prices	subsidize the weaker side monetize the stronger side the winner's own ordering: v_1^k vs v_2^k

Refinement Kills Market-Sharing Equilibrium

For any price $\mathbf{p}$

$$\max_{\mathbf{n}} \Phi_{\mathbf{p}}(\mathbf{n}) = \max\{\Phi_{\mathbf{p}}(\mathbf{A}), \Phi_{\mathbf{p}}(\mathbf{B}), 0\}$$

$\mathbf{A}$ = A-tipping profile (all singlehome on A)

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Market-sharing equilibrium is **not** generically a potential maximizer

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- $\hat{\Phi}_{\mathbf{p}}(\mathbf{n}_1) \equiv \max_{\mathbf{n}_2} \Phi_{\mathbf{p}}(\mathbf{n}_1, \mathbf{n}_2)$ is convex (on the continuously extended domain)
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Stage-1 pricing game is a **winner-take-all Bertrand competition in potential**

Bertrand Competition in Potential: Loser B's Perspective

Stage-1 pricing game is a winner-take-all Bertrand competition in potential:

A-tipping equilibrium is selected $\iff \Phi_p(\mathbf{A}) > \max\{\Phi_p(\mathbf{B}), 0\}$

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Bertrand competition in potential

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B's strongest credible threat:
if B did not maximize it, $\Phi_p(\mathbf{A})$ is below B's max potential, so B would deviate to win the market

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Platform A maximizes its profit given the **potential constraint**:

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- side-1 cost in terms of potential = 1
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 - $v_1^A / v_2^A > 1$: side 2 is more costly $\Rightarrow p_2^A = 0$; side-1 is less costly $\Rightarrow p_1^A > 0$

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The unique equilibrium under potential maximization:

$$\text{A-tipping } (n_1^A = N_1, n_2^A + n_2^{AB} = N_2) \iff \max\{v_1^A, v_2^A\} > \max\{v_1^B, v_2^B\}$$

with equilibrium prices:

$$v_1^A < v_2^A \implies p_1^A < 0 \quad p_2^A > 0; \quad v_1^A > v_2^A \implies p_1^A > 0 \quad p_2^A = 0$$

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with equilibrium prices:

$$v_1^A < v_2^A \implies p_1^A < 0 \quad p_2^A > 0; \quad v_1^A > v_2^A \implies p_1^A > 0 \quad p_2^A = 0$$

$$v_1^B < v_2^B \implies p_1^B < 0 \quad p_2^B > 0; \quad v_1^B > v_2^B \implies p_1^B = 0 \quad p_2^B = 0$$

the winner has higher $\max\{v_1^k, v_2^k\}$
• no tipping to an inferior platform

Theorem: Refined Equilibrium

The unique equilibrium under potential maximization:

$$\text{A-tipping } (n_1^A = N_1, n_2^A + n_2^{AB} = N_2) \iff \max\{v_1^A, v_2^A\} > \max\{v_1^B, v_2^B\}$$

with equilibrium prices:

$$v_1^A < v_2^A \implies p_1^A < 0 \quad p_2^A > 0; \quad v_1^A > v_2^A \implies p_1^A > 0 \quad p_2^A = 0$$

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the winner has higher $\max\{v_1^k, v_2^k\}$

- no tipping to an inferior platform

platforms **subsidize the weaker side** and **monetize the stronger side**

- to earn potential cheaply on the weaker side by subsidies
- to “exchange” potential on the stronger side for revenue

Evans-Schmalensee Puzzle Revisited

	conventional wisdom	Evans-Schmalensee	potential approach
subsidize	singlehoming side	multihoming side	weaker side $\arg \min_i v_i^k$
monetize	multihoming side	singlehoming side	stronger side $\arg \max_i v_i^k$

Generating both observed patterns

- operating systems: $v_{\text{users}}^k > v_{\text{developers}}^k \Rightarrow$ monetize singlehoming side
- game consoles: $v_{\text{users}}^k < v_{\text{developers}}^k \Rightarrow$ monetize multihoming side

Model

Equilibrium Analysis without Refinement

Equilibrium Analysis with Refinement

Platform Design: Exclusive Offers?

jump?

Model

Equilibrium Analysis without Refinement

Equilibrium Analysis with Refinement

Platform Design: Exclusive Offers?

jump?

The **Armstrong-type competitive bottleneck** is a **best-response potential game**

- refinement selects a **unique market-tipping – i.e., winner-take-all – equilibrium**

subsidies	winner	equilibrium pricing
prohibited	higher v_1^k	subsidize the weaker side
permitted	higher $\max\{v_1^k, v_2^k\}$	monetize the stronger side

- market structure is itself a design choice
 - platforms make the relatively favored side exclusive

Appendix

Market-Tipping Equilibrium: Subsidies Permitted

symmetric across two sides

$$\exists \text{ A-tipping equilibrium} \iff v_1^A + v_2^A \geq \max\{v_1^B, v_2^B\}$$

There exist “many” prices that support the market-sharing equilibrium:

- B’s deviation to any divide-and-conquer strategy is unprofitable:

$$p_1^A \leq (v_1^A - \max\{v_1^B, v_2^B\})N_2$$

- A’s profit must $\geq$ A’s profit from deviating to dividing side 1 and conquering side 2:

$$p_1^A N_1 + p_2^A N_2 \geq p_1^B N_1 + (\max\{v_1^A, v_2^A\} - v_1^B)N_1 N_2$$

Market-Sharing Equilibrium: Subsidies Permitted

$$\exists \text{ market-sharing equilibrium} \iff -v_2^A \leq v_1^A - v_1^B \leq v_2^B$$

There exist “many” prices that support the market-sharing equilibrium:

- regularity condition: $p_1^k \leq 0$ and $p_2^k \leq v_2^k n_1^k$
- side-1 indifference condition: $v_1^A N_2 - p_1^A = v_1^B N_2 - p_1^B$
- deviation to any divide-side-1-conquer-side-2 strategy is unprofitable:

$$p_1^A N_1 + p_2^A N_2 \geq p_1^B N_1 + (v_1^B - v_2^A) N_1 N_2$$

$$p_1^B N_1 + p_2^B N_2 \geq p_1^A N_1 + (v_1^A - v_2^B) N_1 N_2$$

Best-Response Potential: Two Definitions

For a finite normal-form game $G = \langle I, (A_i, u_i) \rangle$

- a **Voorneveld best-response potential** $\Phi : A \rightarrow \mathbb{R}$ is such that

$$\arg \max_{a_i} u_i(a_i, a_{-i}) = \arg \max_{a_i} \Phi(a_i, a_{-i}) \quad \forall a_{-i} \in A_{-i}$$

- a **Morris–Ui best-response potential** $\Phi : A \rightarrow \mathbb{R}$ is such that

$$\arg \max_{a_i} u_i(a_i, \lambda_{-i}) = \arg \max_{a_i} \Phi(a_i, \lambda_{-i}) \quad \forall \lambda_{-i} \in \Delta(A_{-i})$$

- ▶ the Voorneveld concept helps **finding** equilibria
- ▶ the Morris–Ui concept helps **refining** equilibria

Theorem: Refined Equilibrium

The unique equilibrium under potential maximization:

$$\text{A-tipping } (n_1^A = N_1, n_2^A + n_2^{AB} = N_2) \iff \max\{v_1^A, v_2^A\} > \max\{v_1^B, v_2^B\}$$

with equilibrium prices:

$$v_1^A < v_2^A: \mathbf{p}^A = (-\max\{v_1^B, v_2^B\} N_2, v_2^A N_1)$$

$$v_1^A > v_2^A: \mathbf{p}^A = ((v_1^A - \max\{v_1^B, v_2^B\})N_2, 0)$$

$$v_1^B < v_2^B: \mathbf{p}^B = (-v_2^B N_2, v_2^B N_1)$$

$$v_1^B > v_2^B: \mathbf{p}^B = (0, 0)$$

How does potential Φ_p change?

How does potential Φ_p change?

- when side-1 agent joins platform k

$$\Phi_p(\mathbf{n}) = \sum_k \left[v_1^k n_1^k (n_2^k + n_2^{AB}) - p_1^k n_1^k - \frac{v_1^k}{v_2^k} p_2^k (n_2^k + n_2^{AB}) \right]$$

$$\Delta\Phi_p = \text{payoff change } v_1^k (n_2^k + n_2^{AB}) - p_1^k$$

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$$\Delta\Phi_p = \text{payoff change } v_1^k (n_2^k + n_2^{AB}) - p_1^k$$

- when side-2 agent joins platform k

$$\Delta\Phi_p = \text{exchange rate } v_1^k / v_2^k \times \text{payoff change } v_2^k n_1^k - p_2^k$$

How does potential Φ_p change?

- when side-1 agent joins platform k

$$\Phi_p(\mathbf{n}) = \sum_k \left[v_1^k n_1^k (n_2^k + n_2^{AB}) - p_1^k n_1^k - \frac{v_1^k}{v_2^k} p_2^k (n_2^k + n_2^{AB}) \right]$$

$$\Delta\Phi_p = \text{payoff change } v_1^k (n_2^k + n_2^{AB}) - p_1^k$$

- when side-2 agent joins platform k

$$\Delta\Phi_p = \text{exchange rate } v_1^k / v_2^k \times \text{payoff change } v_2^k n_1^k - p_2^k$$

- when side-2 agent joins both platforms
 - participation decisions are separable across platforms
 - per-platform sign preservation suffices to preserve best responses

How does potential Φ_p change?

- when side-1 agent joins platform k

$$\Phi_p(\mathbf{n}) = \sum_k \left[v_1^k n_1^k (n_2^k + n_2^{AB}) - p_1^k n_1^k - \frac{v_1^k}{v_2^k} p_2^k (n_2^k + n_2^{AB}) \right]$$

$$\Delta\Phi_p = \text{payoff change } v_1^k (n_2^k + n_2^{AB}) - p_1^k$$

- when side-2 agent joins platform k

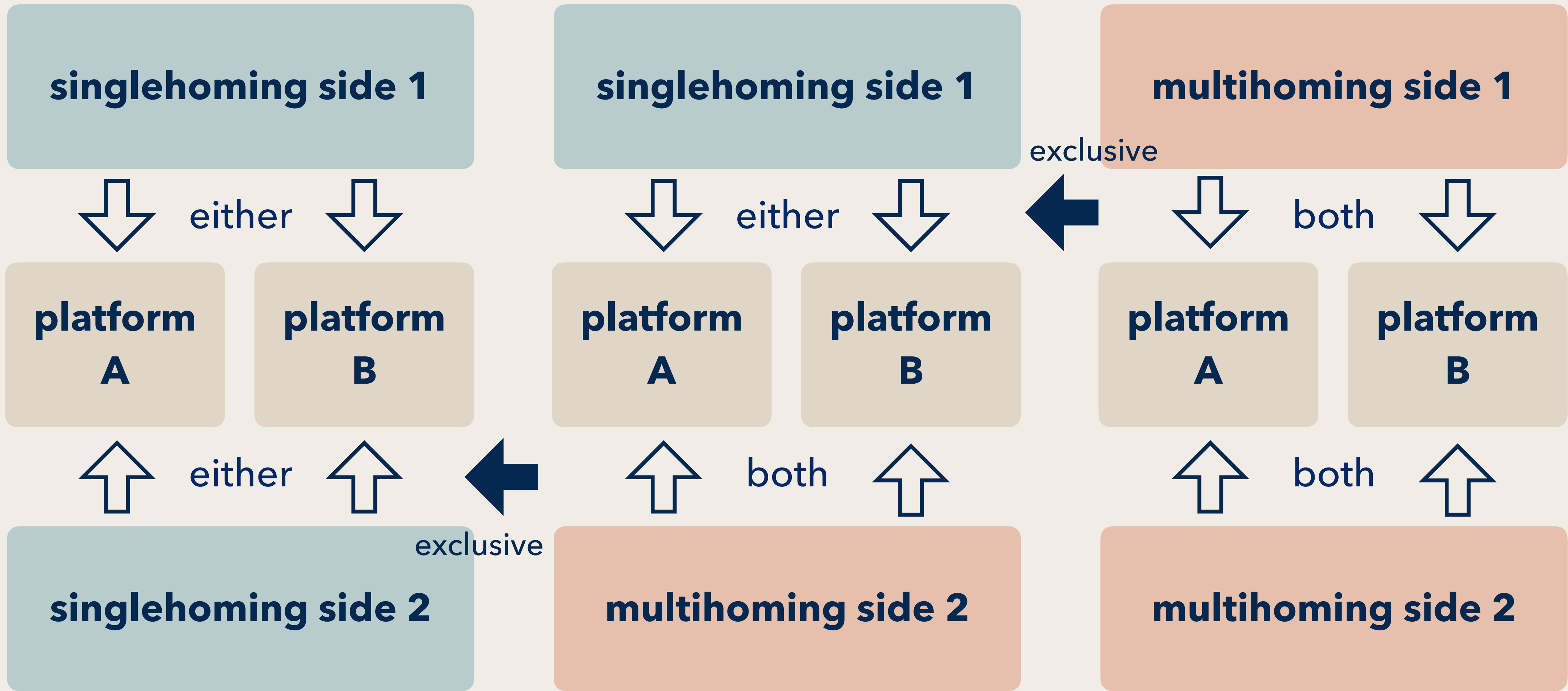
But Φ_p is **not** a potential

- the exchange rate is platform-specific: $v_1^A / v_2^A \neq v_1^B / v_2^B$
- no per-player weight recovers payoff differences

$$\Delta\Phi_p = \text{exchange rate } v_1^k / v_2^k \times \text{payoff change } v_2^k n_1^k - p_2^k$$

- when side-2 agent joins both platforms
 - participation decisions are separable across platforms
 - per-platform sign preservation suffices to preserve best responses

Should Platforms Take Market Structure as Given or Choose It?



Chan 21

Caillaud–Jullien 03; Armstrong–Wright 07

Platform k wins if...

let side-2 agents choose where to interact
when both multihome
⇒ side-1 agents act as singlehomers
≈ competitive bottleneck

subsidies	two-sided singlehoming	competitive bottlenecks	two-sided multihoming
prohibited	higher $v_1^k v_2^k$	higher v_1^k	higher v_1^k
permitted	higher $v_2^k(v_2^k - v_1^{-k})$	higher $\max\{v_1^k, v_2^k\}$	higher $v_1^k + v_2^k$